STAR Lab Video Engagement: Database Schema

One Supabase Postgres database. All project data lives in the **public** schema: four tables and one view. Example rows follow a single made-up study, *Moral Attention Pilot*, in which participant P001 watches three videos in a shuffled order and swipes back once. Real IDs are UUIDs; they are shortened here.

Name	Kind	One row per	Written by
`studies`	table	Study and its experiment settings	Researcher web app
`blocks`	table	Screen in a study (instructions, video, or survey)	Researcher web app
`runs`	table	Participant session on the tablet	Tablet, on sync
`block_views`	table	Time a participant viewed a block	Tablet, on sync
`export_rows`	view	Block view, flattened into the CSV export layout	Nothing (computed from the tables)

How the tables connect

- A study has many `blocks` and many `runs` (each block and run stores `study_id`).
- A run has many `block_views` (each view stores `run_id`).
- Each block view points at the block that was shown (`block_id`). A block can appear in many views.
- `export_rows` joins all four tables into one flat table: one row per block view, with the study name and participant ID repeated as columns. This is what researchers download as a CSV.
- Row Level Security is on for every table with no policies, so only the Python backend (using the secret key) can read or write.
- Studies and blocks that already have participant data cannot be deleted. Deleting a run deletes its block views.

`studies` **(table)**

One row per study a researcher creates, holding the study name and every experiment setting from the web app.

id	name	randomized	show_video_bar	allow_swipe_back	show_response_buttons	allow_pausing	created_at	updated_at
study-1	Moral Attention Pilot	false	true	true	true	false	2026-10-18 09:00	2026-10-18 09:30
study-2	News Clips Study	true	false	false	true	true	2026-10-19 11:15	2026-10-19 11:15

Button customization columns (same rows, continued):

id	left_button_icon_path	right_button_icon_path	left_button_text	right_button_text
study-1	*null*	moral-pilot/heart.png	Dislike	Like
study-2	*null*	*null*	*(blank)*	*(blank)*

What each column holds

- `id`: Unique ID, generated automatically.
- `name`: Study name, shown as Study_ID in the export. Must be unique.
- `randomized`: If true, every block is shuffled. If false, only blocks sharing a random_group are shuffled (see blocks).
- `show_video_bar`: Show each video's progress bar at the top of the screen. Default true.
- `left_button_icon_path`: Where an uploaded left-button image is stored in Supabase Storage. Null uses the default thumbs down.
- `right_button_icon_path`: Where an uploaded right-button image is stored. Null uses the default thumbs up. Here study-1 uses a heart.
- `left_button_text`: Text under the left (thumbs down) button. Default blank.
- `right_button_text`: Text under the right (thumbs up) button. Default blank.
- `allow_swipe_back`: Participant can swipe up to return to the previous block. Default true.
- `show_response_buttons`: Left/right buttons appear over videos. If false they are hidden and inactive. Default true.
- `allow_pausing`: Tapping the screen pauses and resumes the video. Default true.
- `created_at`: When the study was created.
- `updated_at`: When the study was last saved.

`blocks` **(table)**

One row per screen in a study, in the order the researcher placed them. Only the content columns for the block's type are filled in; the others are null.

id	study_id	position	type	label	random_group	body	video_path	file_name	prompt	options	created_at
blk-1	study-1	1	instructions	welcome	*null*	Welcome! Swipe to begin.	*null*	*null*	*null*	*null*	2026-10-18 09:05
blk-2	study-1	2	video	videoA	videos	*null*	moral-pilot/video_a.mp4	video_a.mp4	*null*	*null*	2026-10-18 09:06
blk-3	study-1	3	video	videoB	videos	*null*	moral-pilot/video_b.mp4	video_b.mp4	*null*	*null*	2026-10-18 09:06
blk-4	study-1	4	video	videoC	videos	*null*	moral-pilot/video_c.mp4	video_c.mp4	*null*	*null*	2026-10-18 09:07
blk-5	study-1	5	survey	survey1	*null*	*null*	*null*	*null*	I enjoyed these videos.	{Disagree, Neutral, Agree}	2026-10-18 09:08
blk-6	study-1	6	instructions	goodbye	*null*	Thanks! Swipe to finish.	*null*	*null*	*null*	*null*	2026-10-18 09:08

What each column holds

- `id`: Unique ID for the block.
- `study_id`: The study this block belongs to.
- `position`: The researcher's order, starting at 1. Unique within a study.
- `type`: instructions, video, or survey.
- `label`: Block name used in the export's Block column. Unique within a study.
- `random_group`: Blocks sharing a group are shuffled among the positions they occupy; blocks with null never move. Here positions 2-4 hold the three videos in a random order for each participant.
- `body`: Instructions only: the text shown on screen.
- `video_path`: Video only: where the file is stored in Supabase Storage.
- `file_name`: Video only: the file name, shown as Stimuli in the export.
- `prompt`: Survey only: the question text.
- `options`: Survey only: the answer bubbles, as a list.
- `created_at`: When the block was added.

`runs` **(table)**

One row per participant session on the tablet. The participant ID typed in at the start is the only participant information stored; no names or contact details.

id	study_id	participant_id	started_at	completed_at	synced_at
run-1	study-1	P001	2026-10-20 14:00:00.000	2026-10-20 14:01:20.400	2026-10-20 14:01:22.000
run-2	study-1	P002	2026-10-20 15:30:00.000	*null*	2026-10-20 15:41:05.000

What each column holds

- `id`: Unique ID for the session.
- `study_id`: Which study was run.
- `participant_id`: Entered on the tablet at session start.
- `started_at`: When the session began.
- `completed_at`: When the session ended. Null if it never finished (as for P002).
- `synced_at`: When the tablet uploaded the session. A session is identified by study + participant + start time, so a retried upload cannot create a duplicate.

`block_views` **(table)**

One row per time a participant viewed a block. P001's videos were shuffled to B, C, A; they swiped back from Video A to Video C (row 4), saw C again (row 5), then continued. Returning to a block creates a new row.

id	run_id	block_id	view_order	stimulus	started_at	watch_time_ms	swiped_back	button_press	button_time_ms	video_start_at	video_end_at
1	run-1	blk-1	1	welcome	14:00:05.000	12000	false	*null*	*null*	*null*	*null*
2	run-1	blk-3	2	video_b.mp4	14:00:17.000	15300	false	Right	6100	14:00:17.012	14:00:32.300
3	run-1	blk-4	3	video_c.mp4	14:00:32.300	9800	false	*null*	*null*	14:00:32.315	14:00:42.100
4	run-1	blk-2	4	video_a.mp4	14:00:42.100	3400	**true**	Left	2900	14:00:42.108	14:00:45.500
5	run-1	blk-4	5	video_c.mp4	14:00:45.500	4200	false	*null*	*null*	14:00:45.511	14:00:49.700
6	run-1	blk-2	6	video_a.mp4	14:00:49.700	20000	false	Right	11250	14:00:49.709	14:01:09.700
7	run-1	blk-5	7	survey1	14:01:09.700	7600	false	Agree	5400	*null*	*null*
8	run-1	blk-6	8	goodbye	14:01:17.300	3100	false	*null*	*null*	*null*	*null*

What each column holds

- `id`: Row number, generated automatically.
- `run_id`: The session this view belongs to.
- `block_id`: The block that was shown. blk-4 and blk-2 appear twice because of the swipe back.
- `view_order`: The order things actually happened for this participant (Order in the export). Unique within a run.
- `stimulus`: What was shown (video file name or block label), copied at the time so later edits to the block do not change past data.
- `started_at`: Clock time the block appeared. Each row's start equals the previous start plus its watch time.
- `watch_time_ms`: Milliseconds from the block appearing to the participant swiping away.
- `swiped_back`: True if the participant left this block by swiping back to the previous one.
- `button_press`: Left/Right on videos, the chosen answer on surveys. Null if nothing was pressed.
- `button_time_ms`: Milliseconds from block start to the first button press or survey bubble touch.
- `video_start_at`: Exact clock time video playback began. Null for non-video blocks. Used to sync with physiological recordings.
- `video_end_at`: Exact clock time video playback stopped.

export_rows **(view)**

A saved query, not stored data. It joins all four tables into the flat layout from the design document: one row per block view, with study and participant as columns. Downloading a study's data as a CSV means selecting its rows from this view. Empty button values show as 0. Timestamps are shortened here; real ones include the date.

Study_ID	Participant_ID	Block	Order	Stimuli	Watch_time	Swipe_back	Button_press	Button_time	Video_start_timestamp	Video_end_timestamp
Moral Attention Pilot	P001	welcome	1	welcome	12000	0	0	0	*null*	*null*
Moral Attention Pilot	P001	videoB	2	video_b.mp4	15300	0	Right	6100	14:00:17.012	14:00:32.300
Moral Attention Pilot	P001	videoC	3	video_c.mp4	9800	0	0	0	14:00:32.315	14:00:42.100
Moral Attention Pilot	P001	videoA	4	video_a.mp4	3400	1	Left	2900	14:00:42.108	14:00:45.500
Moral Attention Pilot	P001	videoC	5	video_c.mp4	4200	0	0	0	14:00:45.511	14:00:49.700
Moral Attention Pilot	P001	videoA	6	video_a.mp4	20000	0	Right	11250	14:00:49.709	14:01:09.700
Moral Attention Pilot	P001	survey1	7	survey1	7600	0	Agree	5400	*null*	*null*
Moral Attention Pilot	P001	goodbye	8	goodbye	3100	0	0	0	*null*	*null*

What each column holds

- `Study_ID`: studies.name.
- `Participant_ID`: runs.participant_id.
- `Block`: blocks.label.
- `Order`: block_views.view_order.
- `Stimuli`: block_views.stimulus.
- `Watch_time`: block_views.watch_time_ms, in milliseconds.
- `Swipe_back`: block_views.swiped_back as 1 or 0.
- `Button_press`: block_views.button_press, or 0 if none.
- `Button_time`: block_views.button_time_ms, or 0 if none.
- `Video_start_timestamp`: block_views.video_start_at.
- `Video_end_timestamp`: block_views.video_end_at.
- `run_id, study_id`: Also in the view (not shown) so the export can be filtered to one study or one session.